

Harvest Processing System Implementation

Overview

The system now supports processing harvested birds into inventory batches, meeting all specified requirements for converting live bird counts to processed inventory with yield calculations.

Requirements Implemented

1. Convert live bird counts to processed inventory batches

- **Models:** `HarvestBatch` interface with:
 - `liveBirdsCount`, `averageLiveWeight`, `totalLiveWeight`
 - Automatic calculation of total live weight
 - Batch creation for each cut type
- **API Endpoints:**
 - `POST /api/harvest/process` - Process harvested birds into inventory
 - `GET /api/harvest/batches/:id` - Get harvest batch details
 - `GET /api/harvest/cycles/:cycleId/batches` - Get batches for cycle

2. Create batches for different cuts: whole, breast, thighs, wings, drumsticks

- **Cut Types:** `CutType` enum with:
 - `WHOLE`, `BREAST`, `THIGHS`, `WINGS`, `DRUMSTICKS`, `LEGS`, `GIBLETS`, `FEET`, `BACK`, `NECK`
- **Batch Creation:** Automatic inventory batch creation for each cut
- **Product Mapping:** Each cut type maps to appropriate product subcategory
- **API Endpoint:** `GET /api/harvest/cut-types` - Get available cut types with standard yields

3. Assign batch numbers with harvest date and characteristics

- **Batch Number Generation:** `HARV-YYYYMMDD-001` format
- **Characteristics Tracking:** Harvest date, storage conditions, quality checks
- **Storage Information:** Temperature, humidity, expected shelf life
- **Quality Metrics:** Temperature, pH level, pass/fail results

4. Record weights and storage locations

- **Weight Tracking:** Individual cut weights and total processed weight
- **Storage Locations:** Primary location and specific storage for each cut
- **Quality Checks:** Temperature and pH monitoring
- **Grade Classification:** A, B, C grades for quality assessment

5. Update production cycle status to "completed" upon harvest

- **Cycle Integration:** Links harvest batches to production cycles
- **Status Update:** Automatically updates cycle status to `COMPLETED`
- **Data Synchronization:** Updates cycle with harvest metrics

- **API Integration:** Uses existing `ProductionService` for cycle management

6. Calculate yield percentages

- **Yield Calculations:** `calculateYieldPercentage()` function
- **Waste Calculations:** `calculateWastePercentage()` function
- **Standard Yields:** Pre-defined standard yields for each cut type
- **Variance Analysis:** Comparison of actual vs standard yields
- **API Endpoint:** `GET /api/harvest/batches/:id/yield-analysis` - Detailed yield analysis

Technical Implementation

Backend Structure

1. **Models** (`src/models/HarvestProcessing.ts`):

- `CutType` enum: All chicken cut types
- `HarvestProcessingStatus` enum: PENDING, IN_PROGRESS, COMPLETED, CANCELLED
- Interfaces: `HarvestBatch`, `ProcessHarvestRequest`, `YieldCalculation`, `HarvestAnalytics`
- Utility functions: Yield calculations, batch number generation, product mapping

2. **Service** (`src/services/harvestProcessingService.ts`):

- `HarvestProcessingService` class with static methods
- Batch processing with automatic inventory creation
- Yield analysis and analytics generation
- Integration with production cycles

3. **Routes** (`src/routes/harvest.ts`):

- RESTful API endpoints with authentication
- Comprehensive harvest processing endpoint
- Analytics and reporting endpoints
- Cut type information endpoint

4. **Integration** (`src/app.ts`):

- Added harvest routes to main application
- Updated root endpoint documentation
- Integrated with existing authentication middleware

Key Features

1. Comprehensive Batch Processing

- Automatic calculation of total live weight from bird count and average weight
- Individual batch creation for each cut type
- Quality checks and storage condition tracking
- Grade classification (A, B, C) for quality management

2. Yield Management

- Real-time yield percentage calculations
- Waste tracking and analysis
- Standard yield comparison for quality control
- Variance analysis to identify processing efficiency

3. Production Cycle Integration

- Automatic status update to "completed" upon harvest
- Harvest data synchronization with production cycles
- Maintains existing mortality rate and other cycle metrics
- Links harvest batches to specific production cycles

4. Analytics & Reporting

- Harvest analytics with time-based filtering
- Yield by cut type analysis
- Monthly harvest performance tracking
- Top performing cuts identification
- Dashboard summary for quick insights

5. Inventory Integration

- Automatic inventory batch creation
- Product mapping for each cut type
- Storage location tracking
- Compliance information (halal, antibiotic-free, etc.)

Usage Examples

Processing a Harvest

```
const harvestData = {
  cycleId: 'cycle-123',
  harvestDate: new Date(),
  liveBirdsCount: 1000,
  averageLiveWeight: 2.5,
  cuts: [
    {
      cutType: 'whole',
      weight: 1750,
      count: 1000,
      grade: 'A',
      storageLocation: 'Cold Room A1',
    },
    {
      cutType: 'breast',
```

```

        weight: 625,
        count: 2000,
        grade: 'A',
        storageLocation: 'Cold Room B2',
    },
],
storage: {
    primaryLocation: 'Main Facility',
    temperature: 4,
    humidity: 65,
    expectedShelfLife: 7,
},
};

const result = await apiService.processHarvest(harvestData);

```

Getting Yield Analysis

```

const yieldAnalysis = await apiService.getHarvestYieldAnalysis('batch-123');
console.log(`Overall yield: ${yieldAnalysis.overallYieldPercentage}%`);

```

Getting Harvest Analytics

```

const analytics = await apiService.getHarvestAnalytics();
console.log(`Total birds harvested: ${analytics.totalBirdsHarvested}`);
console.log(`Average yield: ${analytics.averageYieldPercentage}%`);

```

Files Created/Modified

New Files

1. `src/models/HarvestProcessing.ts` - Data models and interfaces
2. `src/services/harvestProcessingService.ts` - Business logic service
3. `src/routes/harvest.ts` - API routes
4. `test-harvest.js` - Test script
5. `HARVEST_PROCESSING_IMPLEMENTATION.md` - This documentation

Modified Files

1. `src/app.ts` - Added harvest routes and updated documentation

API Endpoints

Harvest Processing

- `POST /api/harvest/process` - Process harvested birds into inventory

- `GET /api/harvest/batches/:id` - Get harvest batch details
- `GET /api/harvest/cycles/:cycleId/batches` - Get batches for production cycle

Analytics & Reporting

- `GET /api/harvest/batches/:id/yield-analysis` - Get yield analysis for batch
- `GET /api/harvest/analytics` - Get harvest analytics (with optional date filters)
- `GET /api/harvest/summary` - Get harvest summary for dashboard

Information

- `GET /api/harvest/cut-types` - Get available cut types with standard yields

Testing

A comprehensive test script (`test-harvest.js`) verifies all functionality:

- Live bird count to processed batch conversion
- Multiple cut type batch creation
- Batch number generation with harvest characteristics
- Weight recording and storage location tracking
- Production cycle status updates
- Yield percentage calculations

Next Steps

1. **Frontend UI Components:** Create React components for harvest processing
2. **Real-time Monitoring:** Add real-time yield monitoring during processing
3. **Quality Control Integration:** Integrate with quality control systems
4. **Mobile Support:** Offline harvest recording for field workers
5. **Export Features:** PDF/Excel export for harvest reports
6. **Integration Testing:** End-to-end testing with production data

Conclusion

The harvest processing system has been successfully implemented with all required features. The system provides comprehensive batch processing, yield calculations, production cycle integration, and analytics capabilities, fully meeting the specified requirements for processing harvested birds into inventory.